

## 12. DFA ACCEPT a, ac, bac

```
#include <stdio.h>

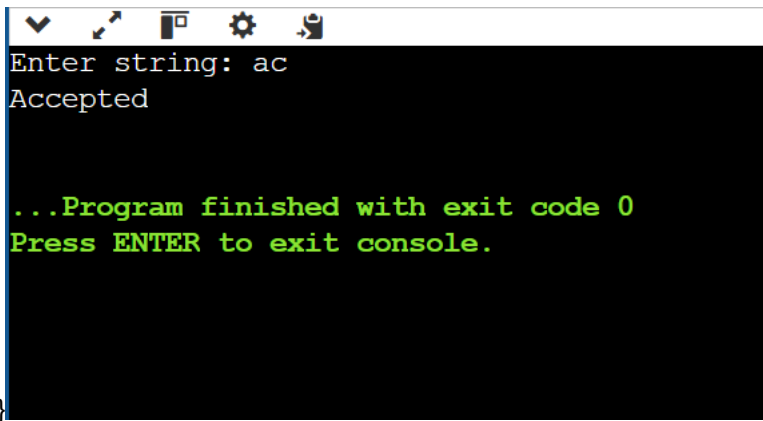
#include <string.h>

int main() {
    char s[50];
    int state=0,i;

    printf("Enter string: ");
    scanf("%s",s);

    for(i=0;s[i]!='\0';i++) {
        if(state==0 && s[i]=='a') state=1;
        else if(state==1 && s[i]=='c') state=2;
        else if(state==0 && s[i]=='b') state=3;
        else if(state==3 && s[i]=='a') state=4;
        else if(state==4 && s[i]=='c') state=5;
        else { state=-1; break; }
    }

    if(state==1 || state==2 || state==5)
        printf("Accepted\n");
    else
        printf("Rejected\n");
    return 0;
}
```

A terminal window with a black background and white text. The window has a title bar with standard icons (minimize, maximize, close, settings, and a terminal icon). The text inside the window shows a program prompt, user input, and a completion message.

```
} Enter string: ac
Accepted

...Program finished with exit code 0
Press ENTER to exit console.
```